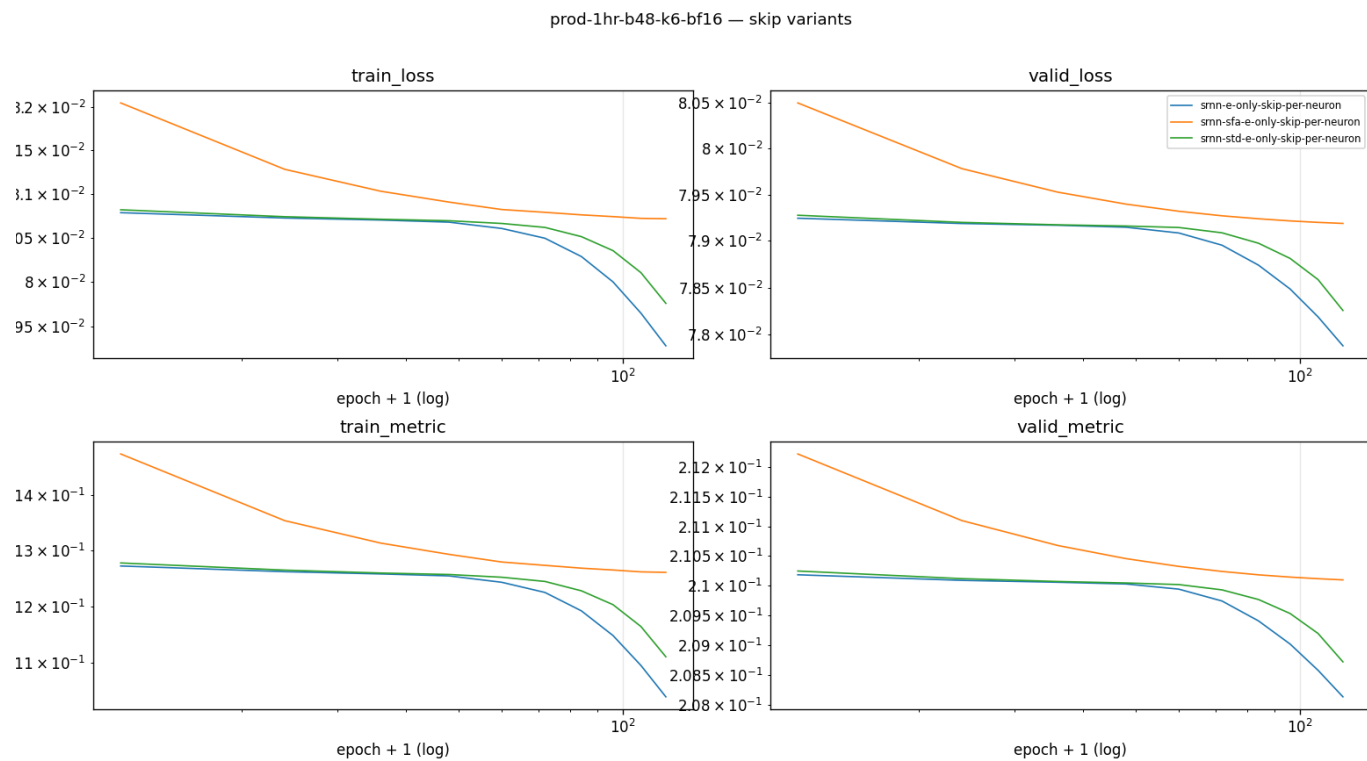


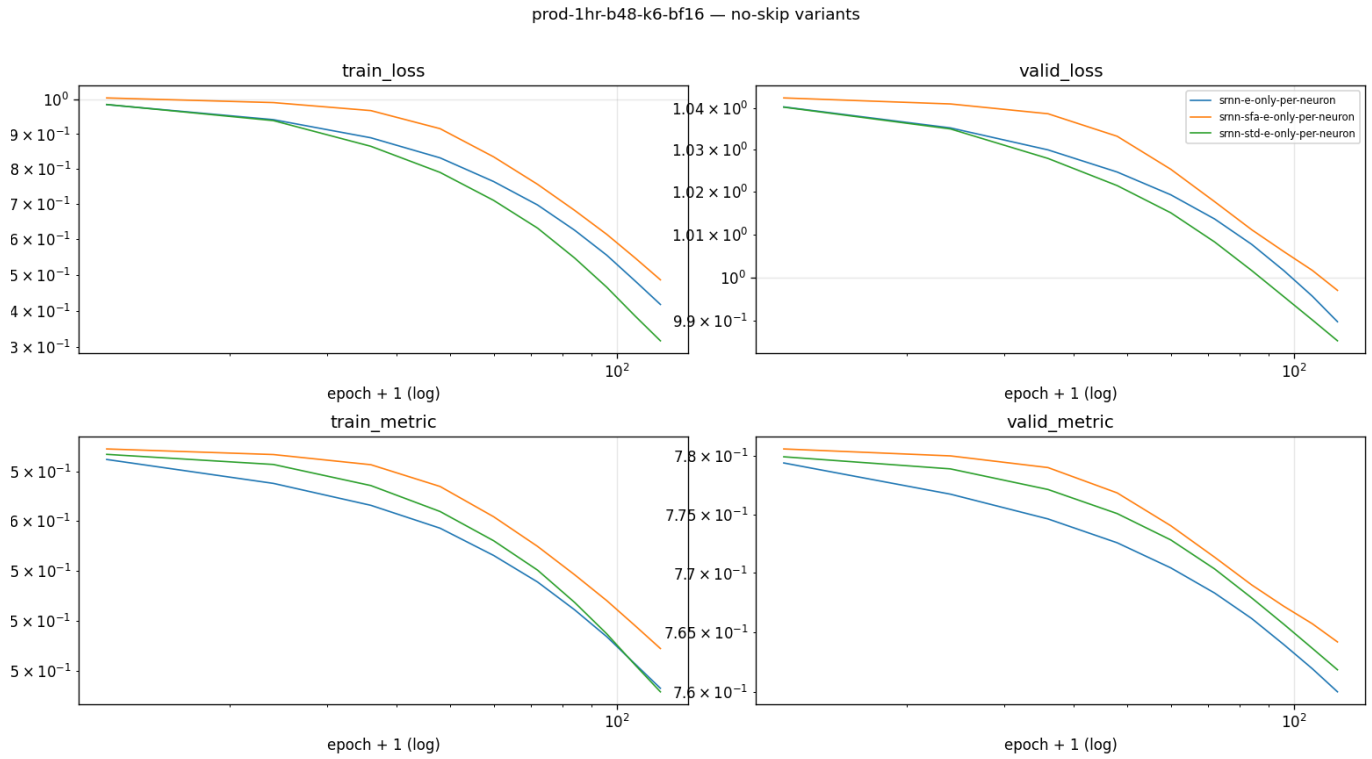
Run: prod-1hr-b48-k6-bf16

Log-log loss curves



Final test_loss (init - last) [skip]:
 srnn-e-only-skip-per-neuron: 0.07877 → 0.07684
 srnn-sfa-e-only-skip-per-neuron: 0.09420 → 0.07809
 srnn-std-e-only-skip-per-neuron: 0.07907 → 0.07727

Figure 1: skip variants — log-log loss/metric



Final test loss (init - last) [no-skip]:
 srnn-e-only-per-neuron: 0.90139 - 0.86651
 srnn-sfa-e-only-per-neuron: 0.91630 - 0.87706
 srnn-std-e-only-per-neuron: 0.90171 - 0.86754

Figure 2: no-skip variants — log-log loss/metric

srnn-e-only-per-neuron

prod-1hr-b48-k6-bf16 — effective time constants (srnn-e-only-per-neuron)

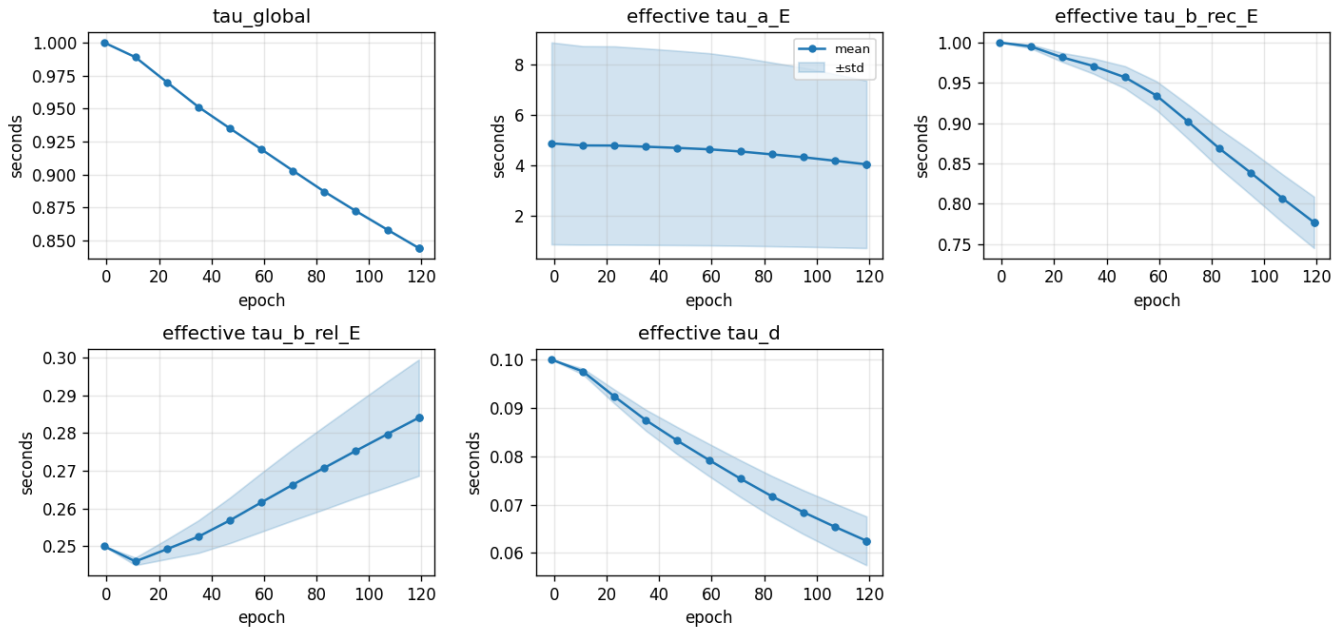
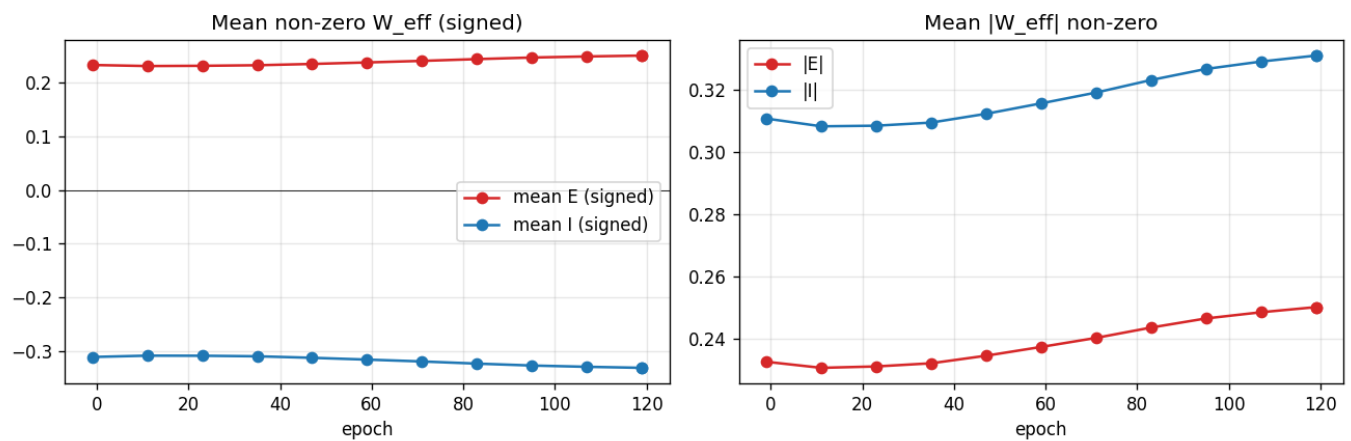


Figure 3: Tau evolution

prod-1hr-b48-k6-bf16 — recurrent E vs I weights (srnn-e-only-per-neuron)

Figure 4: W_{EI} evolution

srnn-e-only-per-neuron — parameter table

prod-1hr-b48-k6-bf16 - variant srnn-e-only-per-neuron (k=0)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.84426	-	-	scalar
tau_d (s)	+0.1	+0.062495	0	0.005043	300
tau_a_E (s)	+4.8777	+4.0495	3.996	3.315	450
tau_b_rec_E (s)	+1	+0.77674	0	0.0319	150
tau_b_rel_E (s)	+0.25	+0.28402	0	0.01541	150
W_eff (E src, signed)	+0.23267	+0.25018	0.07641	0.0832	15048
W_eff (I src, signed)	-0.31062	-0.33081	0.07724	0.08365	14977
W_eff (E src)	+0.23267	+0.25018	0.07641	0.0832	15048
W_eff (I src)	+0.31062	+0.33081	0.07724	0.08365	14977
W_in_eff (input neurons)	-0.0003386	+7.8613e-05	0.1012	0.1224	6675
c_E (SFA coupling)	+0.05	+0.057177	0	0.003967	450
c_0_E (SFA offset)	+0	-0.22391	0	0.05036	450
a_0 (threshold)	+0.35	+0.15498	0	0.06464	300
readout_weight	-2.8854e-05	-0.0032506	0.06604	0.0984	6675
readout_bias	+0	-0.0050603	0	0.0857	89
ic.ic (per-variant init)	+0.10386	+0.10386	0.3828	0.3828	900

srnn-e-only-skip-per-neuron

prod-1hr-b48-k6-bf16 — effective time constants (srnn-e-only-skip-per-neuron)

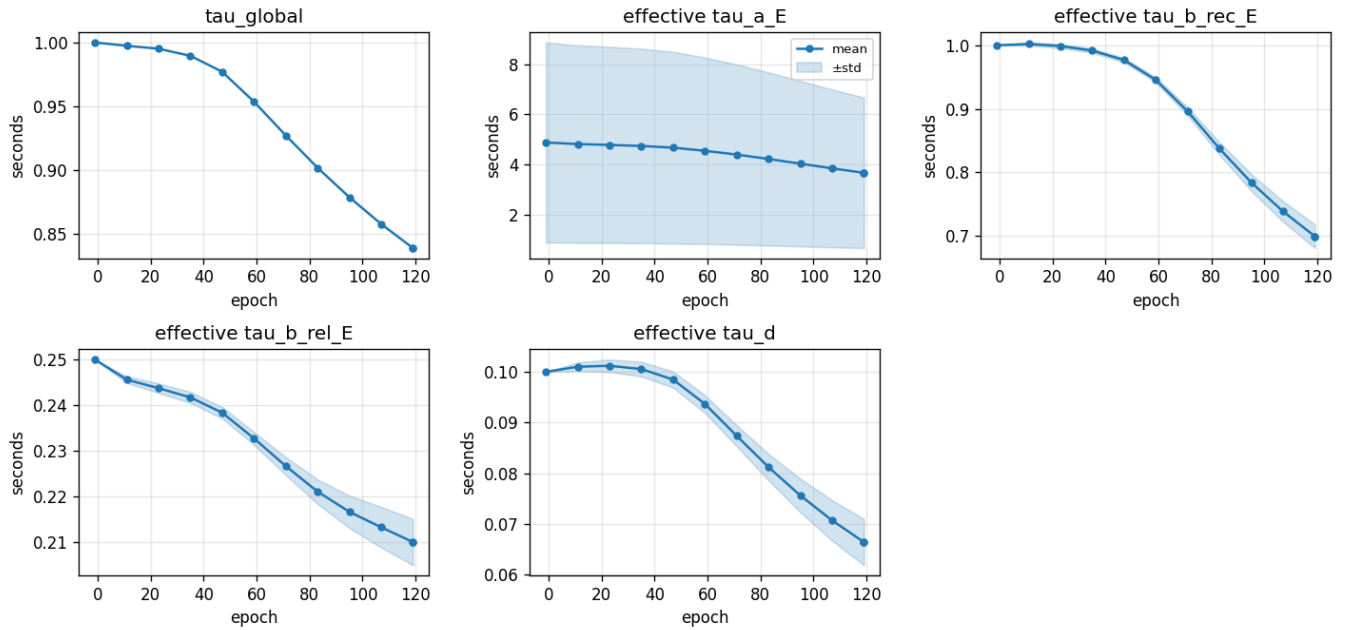
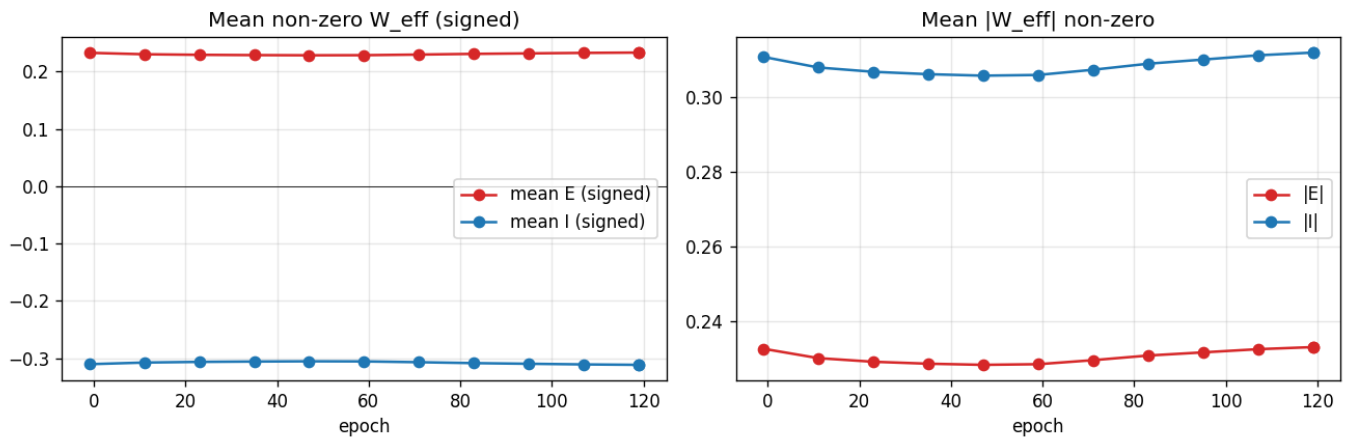


Figure 5: Tau evolution

prod-1hr-b48-k6-bf16 — recurrent E vs I weights (srnn-e-only-skip-per-neuron)

Figure 6: W_{EI} evolution

srnn-e-only-skip-per-neuron — parameter table

```
# prod-1hr-b48-k6-bf16 - variant srnn-e-only-skip-per-neuron (k=1)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.83888	-	-	scalar
tau_d (s)	+0.1	+0.06641	0	0.004549	300
tau_a_E (s)	+4.8777	+3.6655	3.996	3.007	450
tau_b_rec_E (s)	+1	+0.69932	0	0.01853	150
tau_b_rel_E (s)	+0.25	+0.20993	0	0.005072	150
W_eff (E src, signed)	+0.23267	+0.23311	0.07641	0.07676	15048
W_eff (I src, signed)	-0.31062	-0.31177	0.07724	0.07791	14977
W_eff (E src)	+0.23267	+0.23311	0.07641	0.07676	15048
W_eff (I src)	+0.31062	+0.31177	0.07724	0.07791	14977
W_in_eff (input neurons)	-0.00084237	+0.0013641	0.1008	0.08759	6675
c_E (SFA coupling)	+0.05	+0.052384	0	0.001517	450
c_0_E (SFA offset)	+0	+0.014492	0	0.03113	450
a_0 (threshold)	+0.35	+0.33015	0	0.04215	300
readout_weight	+0.00078219	-0.010057	0.06654	0.06103	6675
readout_bias	+0	+0.026863	0	0.03419	89
ic.ic (per-variant init)	+0.10327	+0.10327	0.3829	0.3829	900

srnn-sfa-e-only-per-neuron

prod-1hr-b48-k6-bf16 — effective time constants (srnn-sfa-e-only-per-neuron)

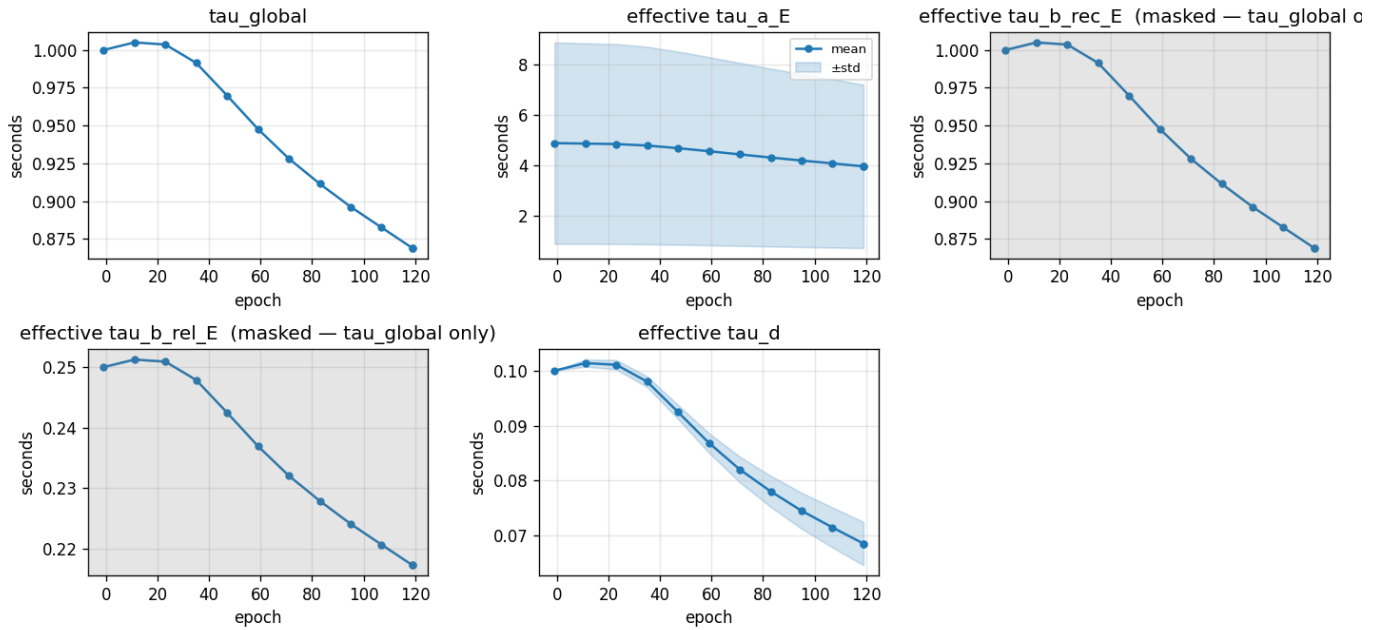
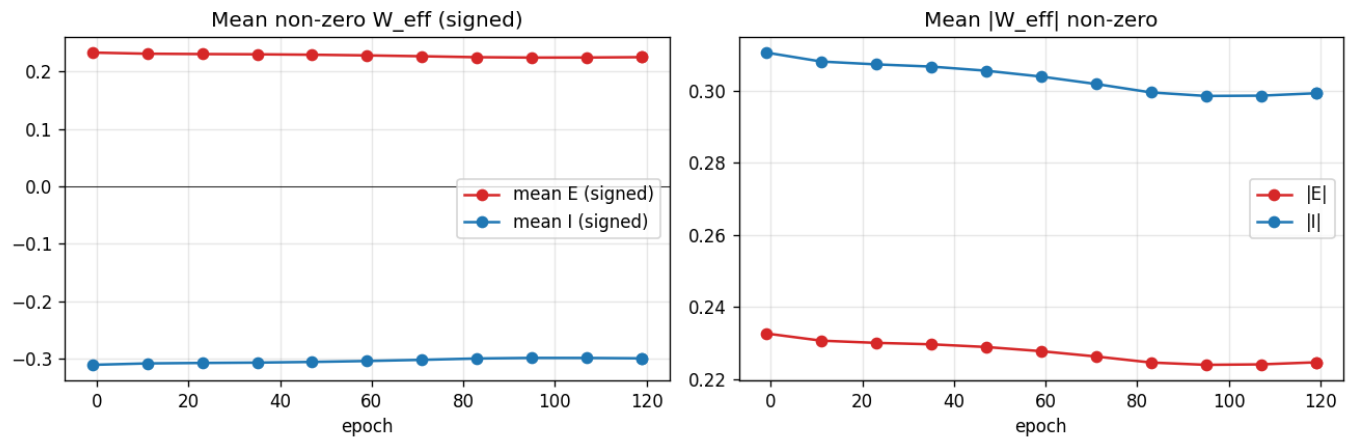


Figure 7: Tau evolution

prod-1hr-b48-k6-bf16 — recurrent E vs I weights (srnn-sfa-e-only-per-neuron)

Figure 8: W_{EI} evolution

srnn-sfa-e-only-per-neuron — parameter table

```
# prod-1hr-b48-k6-bf16 - variant srnn-sfa-e-only-per-neuron (k=2)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.86885	-	-	scalar
tau_d (s)	+0.1	+0.068555	0	0.003936	300
tau_a_E (s)	+4.8777	+3.9551	3.996	3.244	450
tau_b_rec_E (s)	+1	+0.86885	0	0	150
tau_b_rel_E (s)	+0.25	+0.21721	0	0	150
W_eff (E src, signed)	+0.23267	+0.22467	0.07641	0.07414	15048
W_eff (I src, signed)	-0.31062	-0.29933	0.07724	0.07477	14977
W_eff (E src)	+0.23267	+0.22467	0.07641	0.07414	15048
W_eff (I src)	+0.31062	+0.29933	0.07724	0.07477	14977
W_in_eff (input neurons)	-0.00015382	-2.4072e-05	0.09934	0.1176	6675
c_E (SFA coupling)	+0.05	+0.051522	0	0.001539	450
c_0_E (SFA offset)	+0	-0.027567	0	0.03823	450
a_0 (threshold)	+0.35	+0.34634	0	0.0325	300
readout_weight	-0.00098972	+0.00088098	0.06678	0.0803	6675
readout_bias	+0	-0.0010009	0	0.02886	89
ic.ic (per-variant init)	+0.13078	+0.13078	0.4172	0.4172	900

srnn-sfa-e-only-skip-per-neuron

prod-1hr-b48-k6-bf16 — effective time constants (srnn-sfa-e-only-skip-per-neuron)

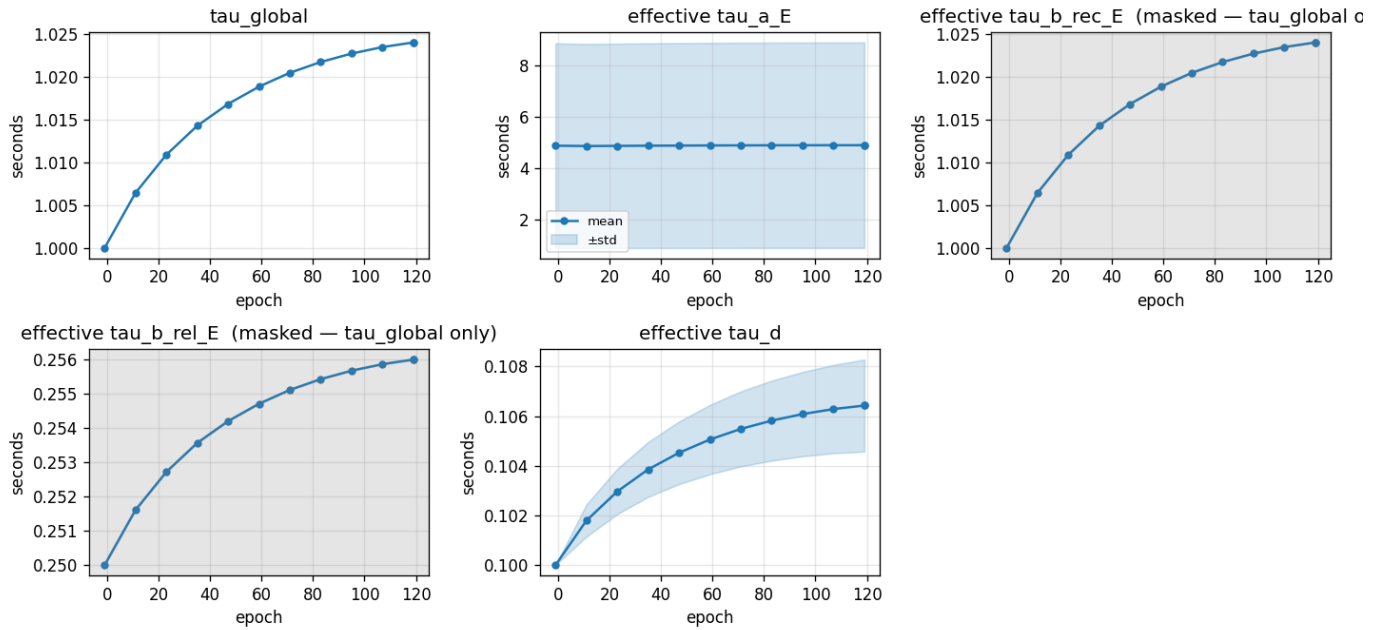
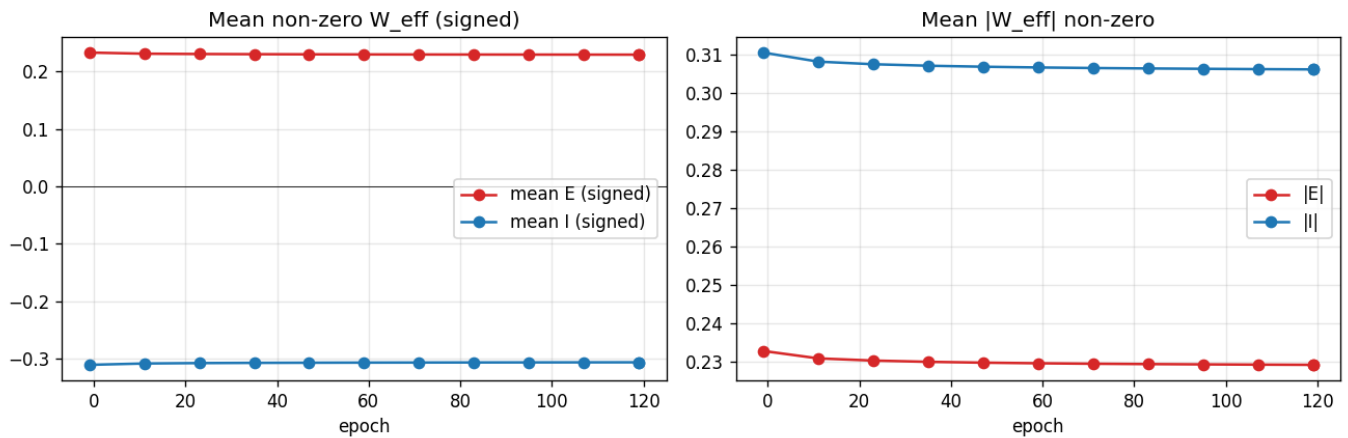


Figure 9: Tau evolution

prod-1hr-b48-k6-bf16 — recurrent E vs I weights (srnn-sfa-e-only-skip-per-neuron)

Figure 10: W_{EI} evolution

srnn-sfa-e-only-skip-per-neuron — parameter table

```
# prod-1hr-b48-k6-bf16 - variant srnn-sfa-e-only-skip-per-neuron (k=3)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+1.024	-	-	scalar
tau_d (s)	+0.1	+0.10643	0	0.001852	300
tau_a_E (s)	+4.8777	+4.8957	3.996	4.014	450
tau_b_rec_E (s)	+1	+1.024	0	0	150
tau_b_rel_E (s)	+0.25	+0.25599	0	0	150
W_eff (E src, signed)	+0.23267	+0.22905	0.07641	0.07524	15048
W_eff (I src, signed)	-0.31062	-0.30623	0.07724	0.07617	14977
W_eff (E src)	+0.23267	+0.22905	0.07641	0.07524	15048
W_eff (I src)	+0.31062	+0.30623	0.07724	0.07617	14977
W_in_eff (input neurons)	-0.0011621	-0.00078498	0.09962	0.09376	6675
c_E (SFA coupling)	+0.05	+0.05166	0	0.0006848	450
c_0_E (SFA offset)	+0	+0.040759	0	0.01187	450
a_0 (threshold)	+0.35	+0.3793	0	0.02108	300
readout_weight	+0.00016801	-0.00022955	0.0673	0.06042	6675
readout_bias	+0	-0.00013023	0	0.00185	89
ic.ic (per-variant init)	+0.09231	+0.09231	0.5294	0.5294	900

srnn-std-e-only-per-neuron

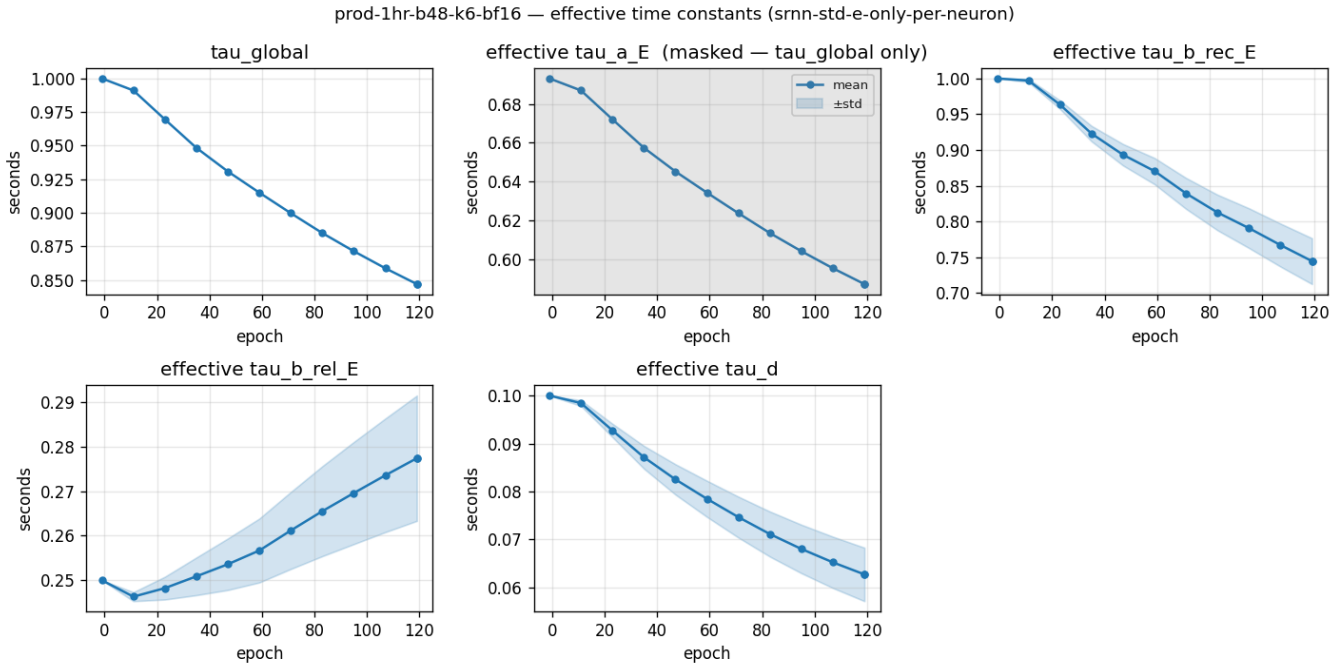
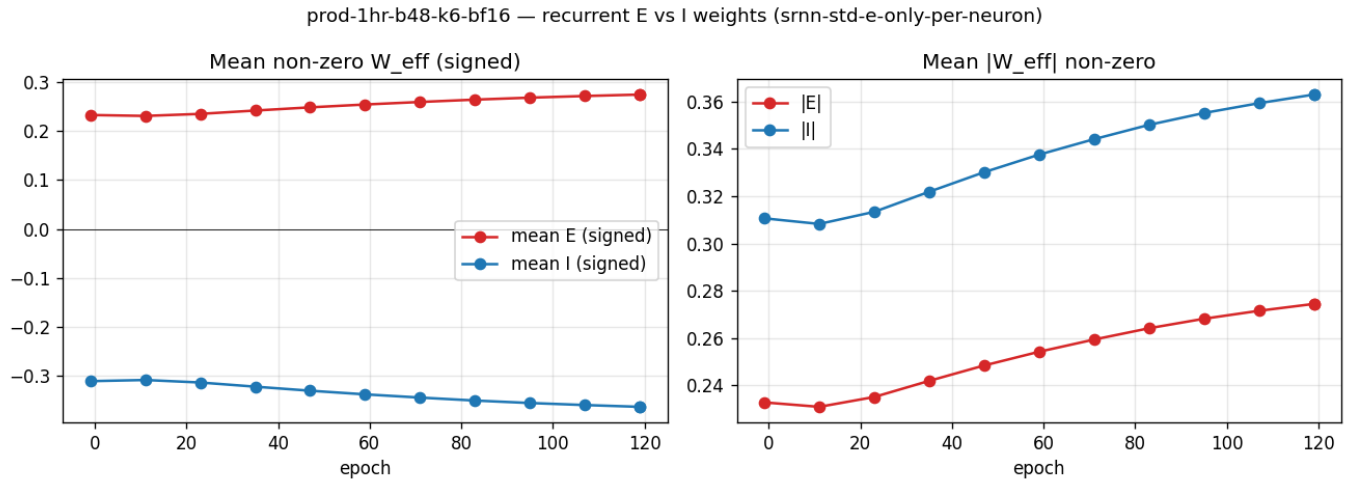


Figure 11: Tau evolution

Figure 12: W_{EI} evolution

srnn-std-e-only-per-neuron — parameter table

```
# prod-1hr-b48-k6-bf16 - variant srnn-std-e-only-per-neuron (k=4)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.84706	-	-	scalar
tau_d (s)	+0.1	+0.062714	0	0.005576	300
tau_a_E (s)	+0.69315	+0.58714	0	0	450
tau_b_rec_E (s)	+1	+0.74452	0	0.03208	150
tau_b_rel_E (s)	+0.25	+0.27741	0	0.01408	150
W_eff (E src, signed)	+0.23267	+0.27438	0.07641	0.09116	15048
W_eff (I src, signed)	-0.31062	-0.36304	0.07724	0.0915	14977
W_eff (E src)	+0.23267	+0.27438	0.07641	0.09116	15048
W_eff (I src)	+0.31062	+0.36304	0.07724	0.0915	14977
W_in_eff (input neurons)	+0.00019944	-0.00017918	0.09994	0.1175	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.18151	0	0.05688	300
readout_weight	+0.00074661	+0.0080122	0.06657	0.09976	6675
readout_bias	+0	-0.01502	0	0.09724	89
ic.ic (per-variant init)	+0.090819	+0.090819	0.3835	0.3835	900

srnn-std-e-only-skip-per-neuron

prod-1hr-b48-k6-bf16 — effective time constants (srnn-std-e-only-skip-per-neuron)

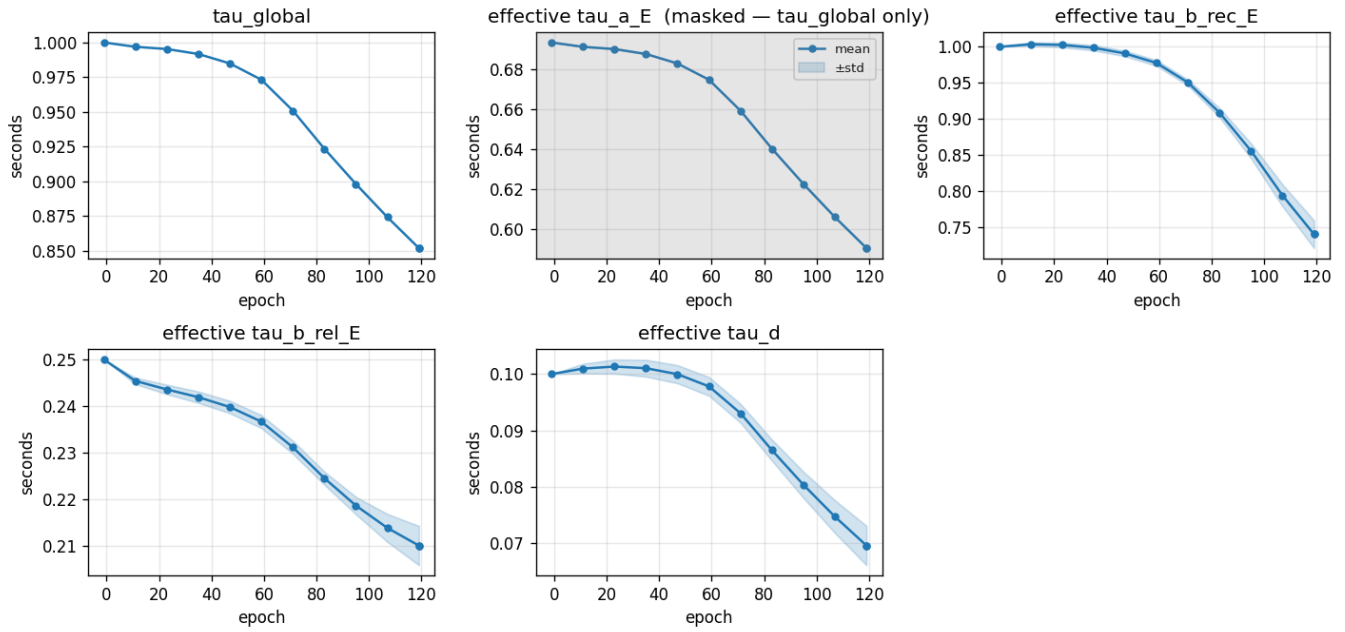


Figure 13: Tau evolution

prod-1hr-b48-k6-bf16 — recurrent E vs I weights (srnn-std-e-only-skip-per-neuron)

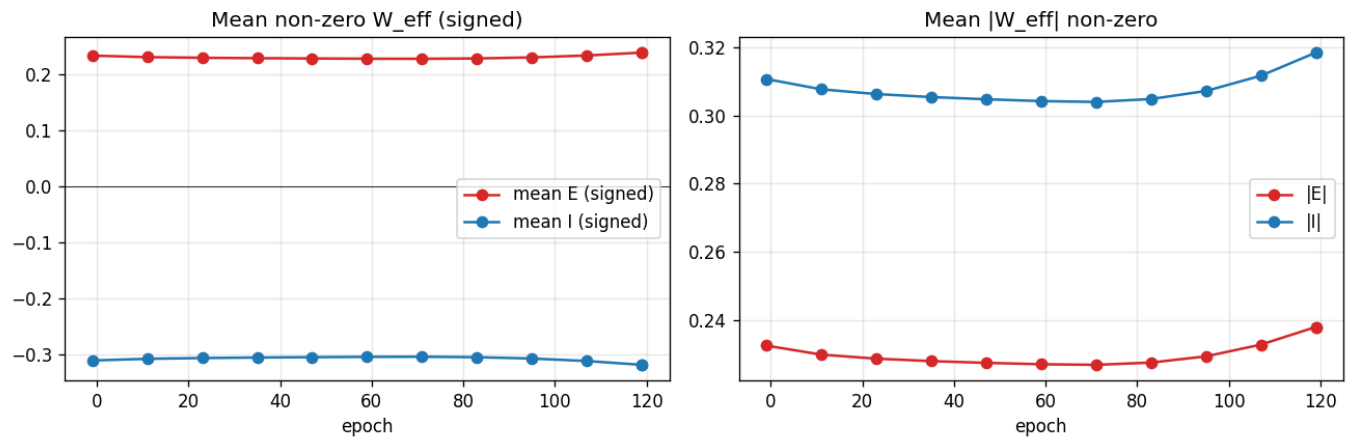


Figure 14: W_EI evolution

srnn-std-e-only-skip-per-neuron — parameter table

prod-1hr-b48-k6-bf16 - variant srnn-std-e-only-skip-per-neuron (k=5)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.85207	-	-	scalar
tau_d (s)	+0.1	+0.069634	0	0.003489	300
tau_a_E (s)	+0.69315	+0.59061	0	0	450
tau_b_rec_E (s)	+1	+0.74039	0	0.01912	150
tau_b_rel_E (s)	+0.25	+0.21009	0	0.004211	150
W_eff (E src, signed)	+0.23267	+0.23812	0.07641	0.07839	15048
W_eff (I src, signed)	-0.31062	-0.31836	0.07724	0.07932	14977
W_eff (E src)	+0.23267	+0.23812	0.07641	0.07839	15048
W_eff (I src)	+0.31062	+0.31836	0.07724	0.07932	14977
W_in_eff (input neurons)	+0.00036119	-0.00043907	0.09956	0.095	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.36411	0	0.0305	300
readout_weight	+0.00055372	-0.0019413	0.06572	0.06001	6675
readout_bias	+0	+0.00020188	0	0.02103	89
ic.ic (per-variant init)	+0.083543	+0.083543	0.384	0.384	900